COMP 3613 Assignment 2

Individual report

Student Awards Testing Submission

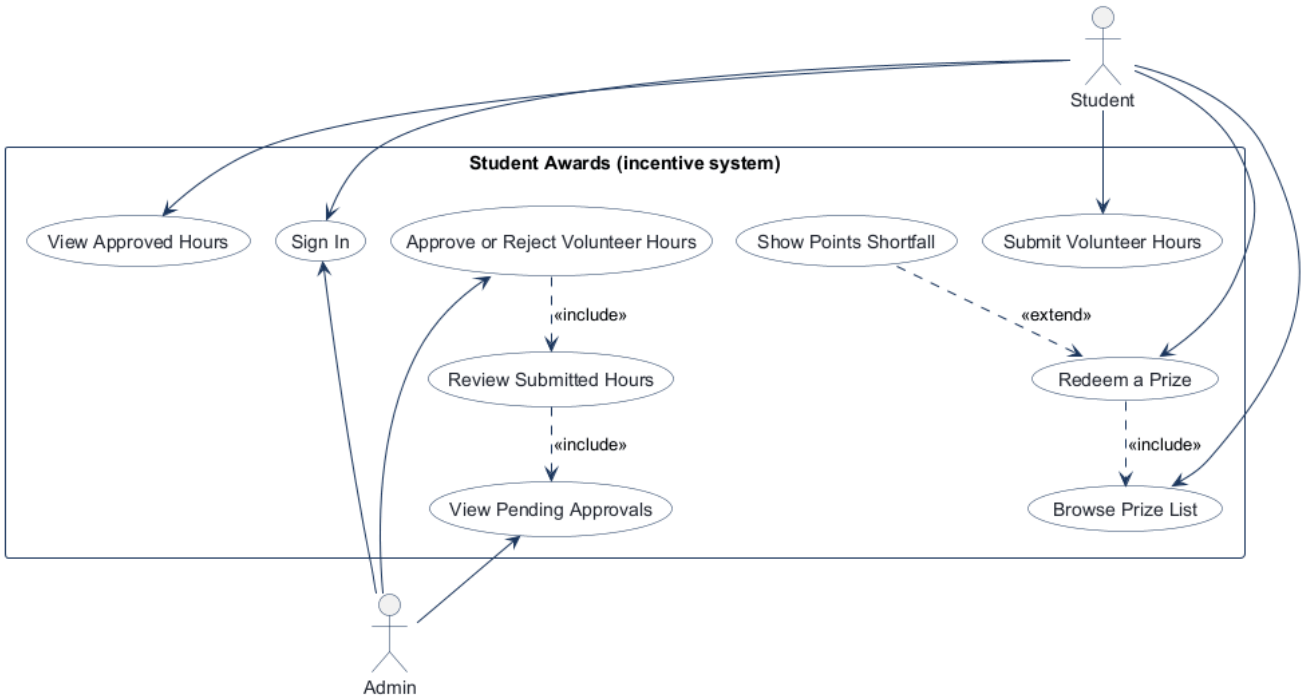
Project testing deliverables: plans, implementation summary, user acceptance testing, automated test reports, and Postman API evidence.

Student	Details
Name	Aliyah Ruan
Student ID	816044540

Project Description

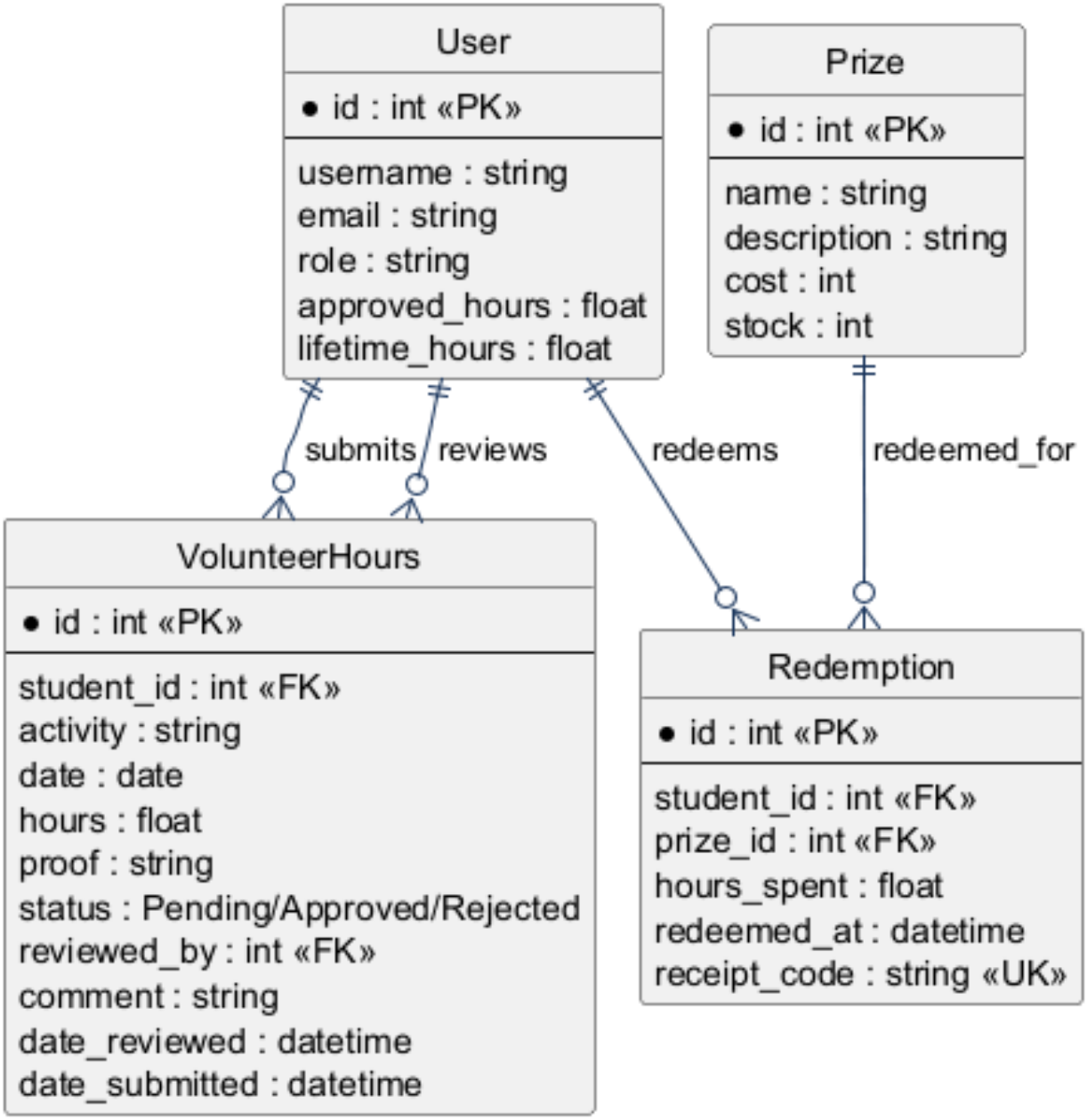
Student Awards is a FastAPI incentive application for tracking student volunteer hours and exchanging approved hours for prizes. Students submit volunteer activities with proof for admin review. Approved submissions increase the student's spendable approved-hours balance and cumulative lifetime-hours total. Students can browse prizes, see stock and point requirements, and redeem an eligible prize with a receipt and QR code. Admins can approve or reject pending submissions; a rejection requires a comment.

Use Case Diagram



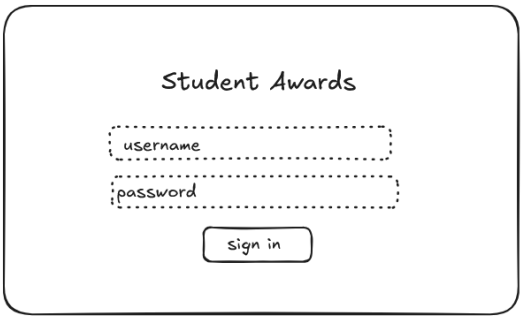
The Student actor signs in, submits hours, views approved hours, browses prizes, and redeems prizes. The Admin actor signs in, views pending approvals, and reviews submissions for approval or rejection. Prize redemption includes browsing the prize list; a points-shortfall message is a conditional extension.

Entity Relationship Diagram

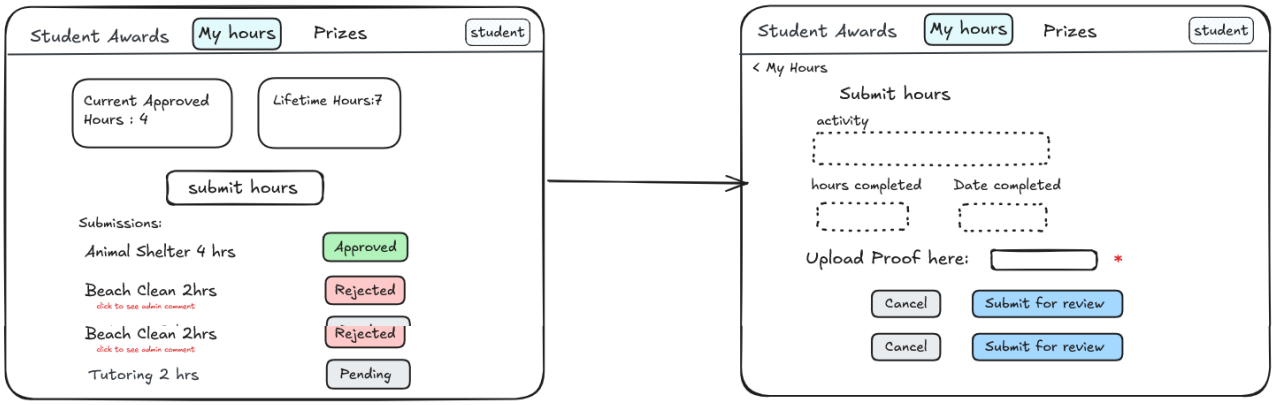


reviewed_by refers to the admin User who reviewed the submission. A Redemption links a student to a prize and stores the hours spent and unique receipt code.

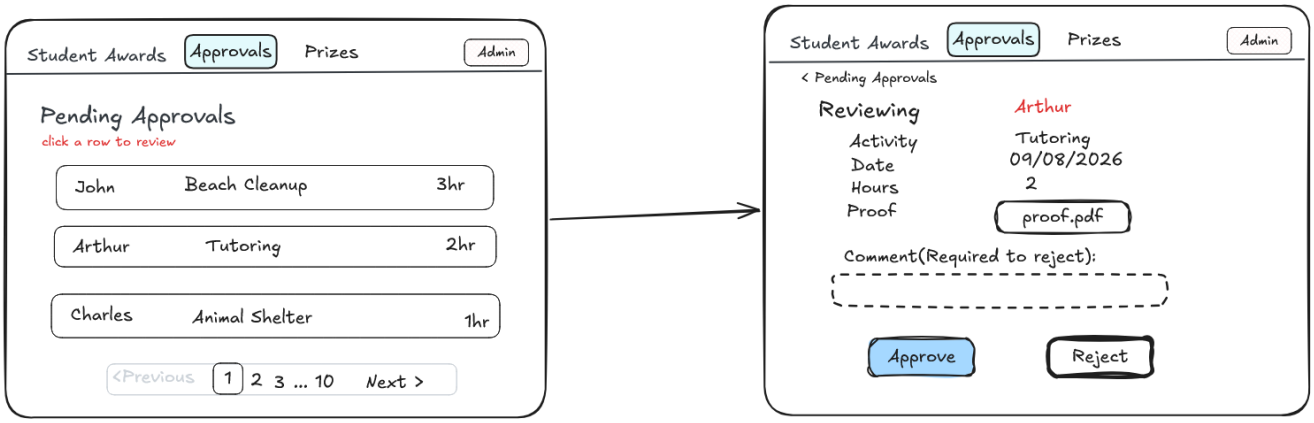
Application Wireframe



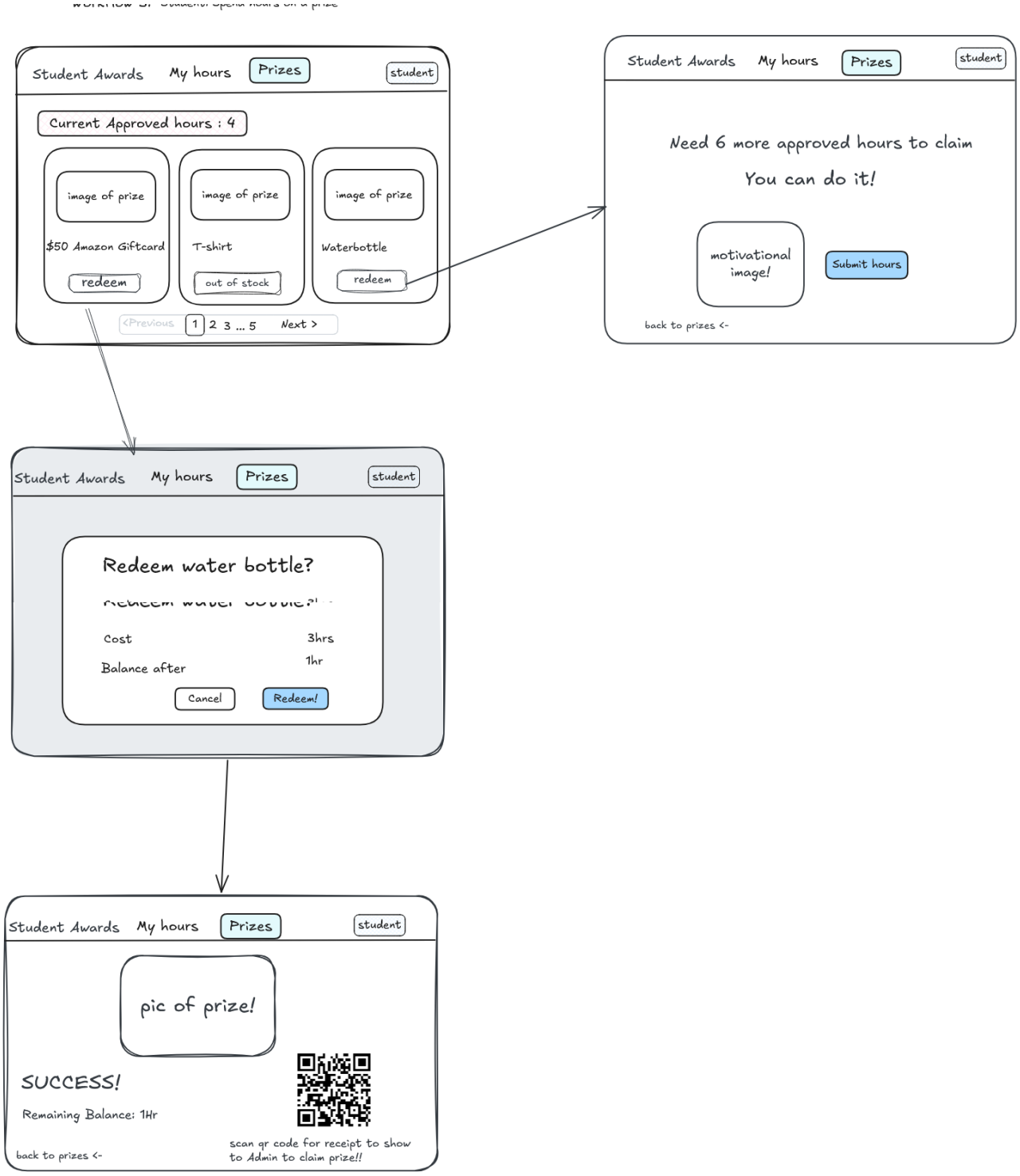
workflow 1: Student: Log volunteer hours, my hours and submit hours



workflow 2: Admin, approve or reject submitted hours



Workflow 3: Student: Spend hours on a prize



Test Plans

The test plans and UAT cases follow the wireframe workflows:

1. Submit volunteer hours as a student.
2. Approve or reject a submission as an admin.
3. Browse prizes and redeem one as a student.

User Acceptance Test Record

I manually executed the following UAT cases through the Student Awards application using the student and administrator accounts. I checked each result against the acceptance criteria and recorded the outcome below.

Test setup: Student account bob; administrator account admin. Before the captured approval, Bob had 2.0 approved hours and 12.0 lifetime hours. The prize-shop scenarios used a 6.0-hour approved balance. The catalog showed the T-shirt out of stock, the $50 giftcard at 10 hours, and the Waterbottle at 3 hours.

ID	Preconditions and test action	Acceptance criteria	Actual result, evidence, and status
UAT-01	Sign in as student bob.	Student session opens with My Hours and Prizes navigation.	The student dashboard displayed My Hours and Prizes navigation. Pass.
UAT-02	As bob, submit 4 hours for Animal Shelter with date and proof.	Submission is Pending; balance remains 2.0 before review.	The submission showed Pending and Bob's balance remained 2.0. Pass.
UAT-03	As admin, approve Bob's pending 4-hour Animal Shelter submission.	Status becomes Approved; approved/lifetime totals increase by 4.0.	The submission became Approved; balances changed 2.0 -> 6.0 and 12.0 -> 16.0. Pass.
UAT-04	As admin, reject a pending submission with comment blank.	Rejection is blocked and required-comment validation appears.	The application displayed "A comment is required when rejecting a submission." Pass.
UAT-05	As admin, reject Bob's 2-hour Planting trees submission with "proof is blurry".	Rejection completes with the supplied comment.	I entered "proof is blurry" and completed the rejection. Pass.
UAT-06	As bob with 6.0 approved hours, view the T-shirt in Prizes.	Balance is 6.0 and T-shirt availability is shown as out of stock.	The 6.0 balance and "Out of stock" label were displayed. Pass.

ID	Preconditions and test action	Acceptance criteria	Actual result, evidence, and status
UAT-07	As bob, view the $50 giftcard costing 10 hours.	Shortfall of 4.0 hours is stated; submit-hours and return options are available.	The page stated that 4.0 more hours were needed and showed both options. Pass.
UAT-08	As bob, select the 3-hour Waterbottle and cancel confirmation.	Cancellation returns to Prizes without redeeming.	The confirmation showed the 3-hour cost and projected balance; cancel returned to Prizes without redeeming. Pass.
UAT-09	As bob, confirm the Waterbottle redemption.	Success shows 3.0 remaining hours and receipt/QR.	The success page showed a 3.0 balance, QR code, and receipt code. Pass.

Unit Test Plan

Unit tests isolate service behaviour using a test database and controlled student, admin, and prize records.

ID	Test case	Expected result
UT-01	Submit volunteer hours with a value of zero.	VolunteerHoursService raises a validation error stating hours must be more than zero.
UT-02	Approve a pending 3.5-hour submission.	The submission becomes Approved, and the student's approved and lifetime totals each increase by 3.5 hours.
UT-03	Redeem a 4-hour prize with 10 approved hours and stock of 2.	A redemption records 4 hours spent, the student's balance becomes 6 hours, and prize stock becomes 1.

Integration Test Plan

FastAPI TestClient tests exercise complete HTTP requests against an isolated test database, including authentication, request validation, route responses, and persisted state.

ID	Test case	Expected result
IT-01	Submit volunteer hours through the browser form with a proof upload.	The route redirects with status 303 and stores one Pending submission for the student.
IT-02	Create a volunteer-hours resource through POST /api/hours with form fields and a proof upload.	The route returns 201 with the public resource fields and does not expose the stored proof path.
IT-03	Submit zero hours through the browser form.	Request validation returns 422 and no valid submission is created.
IT-04	Approve a student's pending submission as an admin.	The route redirects with status 303, the submission becomes Approved, and the student's approved balance increases.
IT-05	Redeem a prize as an authenticated student with sufficient hours.	The route redirects with status 303 and prize stock decreases by one.
IT-06	Request the admin page without an authenticated session.	The route returns 401 Unauthorized.
IT-07	Request the admin page as an authenticated student.	The route returns 403 Forbidden.

Run the automated plans from the repository root with:

```
.\.venv\Scripts\python.exe -m pytest tests\test_services.py tests\test_routes.py -q
```

Reference documents supplied for this assignment:

- COMP 3613 Test Plans
- User Acceptance Test Plan reference

Unit Tests

Test file: tests/test_services.py

Command:

```
.\.venv\Scripts\python.exe -m pytest tests\test_services.py -q
```

Result: **3 passed, 1 non-failing deprecation warning.**

Integration Tests

Test file: tests/test_routes.py

Command:

```
.\.venv\Scripts\python.exe -m pytest tests\test_routes.py -q
```

Result: **7 passed, 1 non-failing deprecation warning.**

API Tests (Postman)

The exported collection is student-awards-api.postman_collection.json. It exercises the actual FastAPI routes and cookie-based authentication, including invalid-hours validation, authorization, both form and JSON resource submissions, review, redemption, receipt, and QR responses. The latest Newman run completed with **19 assertions passed and 0 failed** against the freshly seeded disposable database at http://127.0.0.1:5000.

Implementation

The application uses FastAPI and SQLModel with a layered structure: **routes -> services -> repositories -> models**. Routes bind form/API input and delegate business operations to services; repositories handle persistence. Key implementation files: hours API route, hours response schema, unit tests, and integration tests. Student routes provide hours history, proof upload, prize browsing, shortfall guidance, prize confirmation, redemption, and QR receipts. Admin routes provide a pending-approval list and submission review. The review service enforces Pending-only decisions and requires a comment to reject. Prize redemption checks stock and approved balance before updating the balance and stock.

Repository

https://github.com/aliyahrr/comp3613a1

Test Reports

Unit Test Report

student-awards-unit-report.html

Report generated on 10-Nov-2026 at 16:55:34 by pytest-html v4.2.0

Environment

Python	3.14.5
Platform	Windows-11-10.0.26300-SP0
Packages	• pytest: 9.1.1 • pluggy: 1.6.0
Plugins	• anyio: 4.15.1 • asyncio: 1.4.0 • html: 4.2.0 • metadata: 3.1.1

Summary

3 tests took 276 ms.

(Un)check the boxes to filter the results.

0 Failed, 3 Passed, 0 Skipped, 0 Expected failures, 0 Unexpected passes, 0 Errors, 0 Reruns 0 Retried, Show all details / Hide all details

Result	Test	Duration	Links
Passed	tests/test_services.py::test_submit_hours_rejects_invalid_hours	65 ms	
Passed	tests/test_services.py::test_review_submission_approves_hours_and_updates_totals	112 ms	
Passed	tests/test_services.py::test_redeem_prize_updates_stock_and_balance	66 ms	

Integration Test Report

student-awards-integration-report.html

Report generated on 10-Nov-2026 at 16:55:35 by pytest-html v4.2.0

Environment

Python	3.14.5
Platform	Windows-11-10.0.26300-SP0
Packages	• pytest: 9.1.1 • pluggy: 1.6.0
Plugins	• anyio: 4.15.1 • asyncio: 1.4.0 • html: 4.2.0 • metadata: 3.1.1

Summary

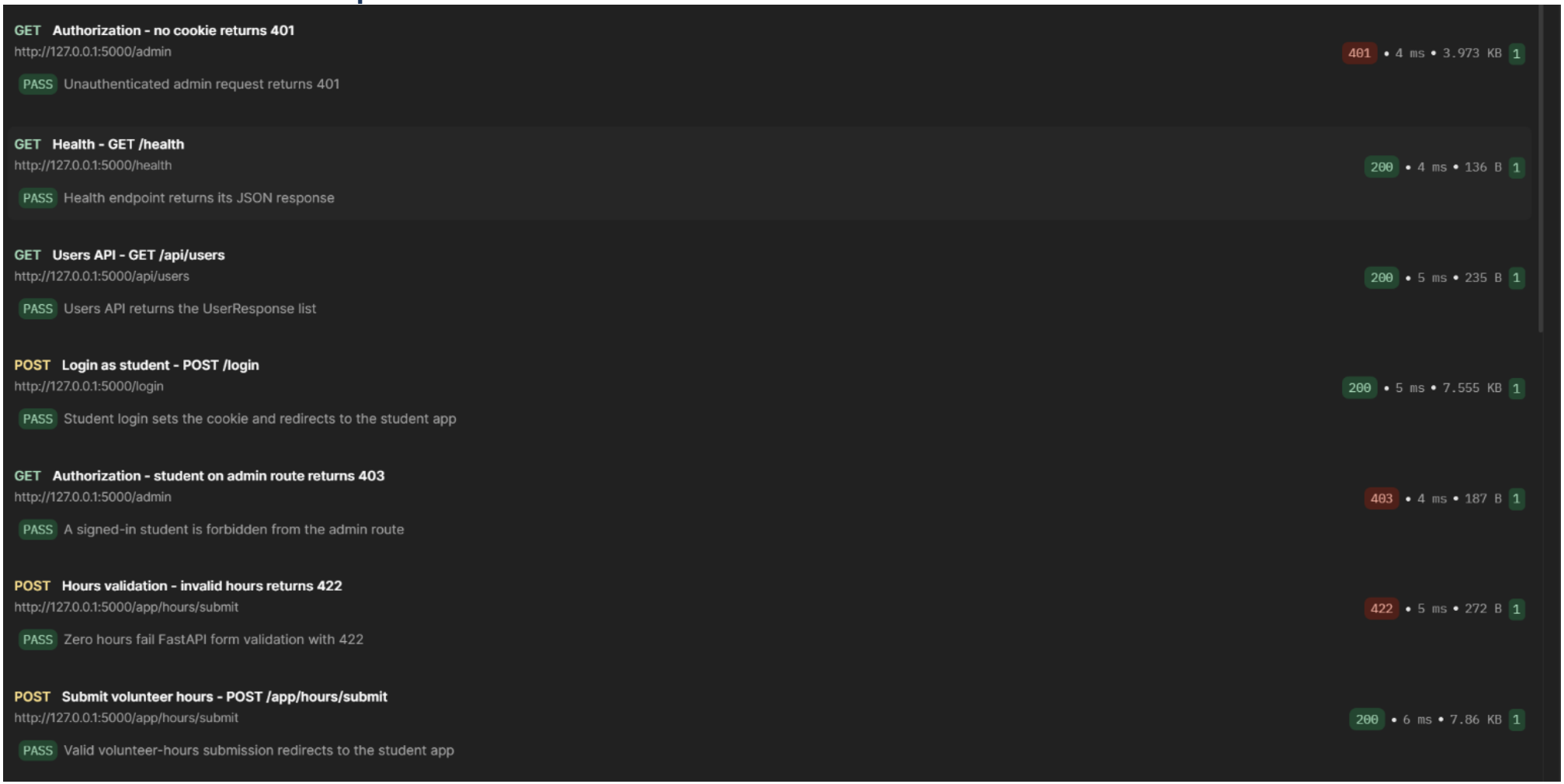
7 tests took 544 ms.

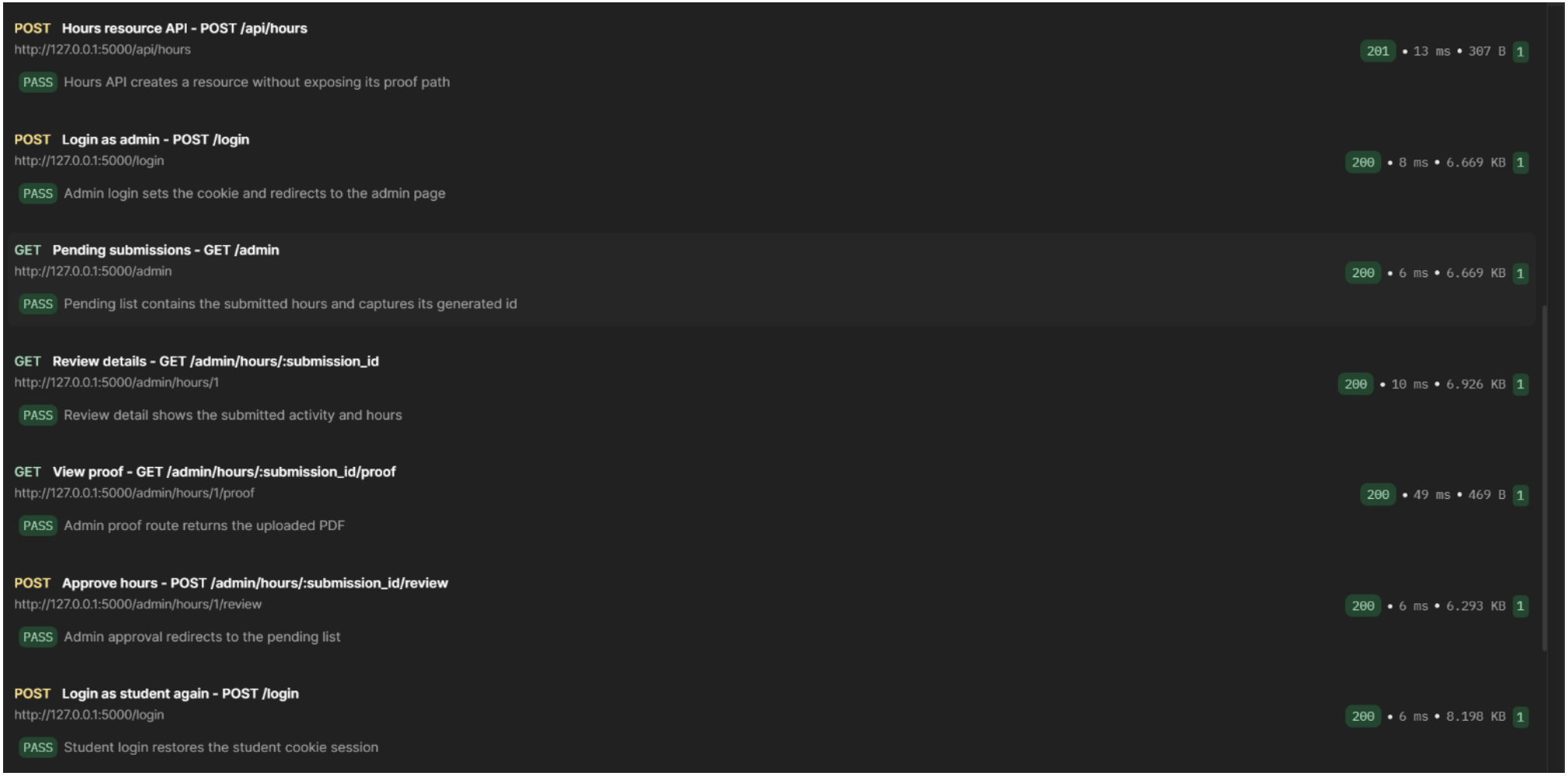
(Un)check the boxes to filter the results.

0 Failed, 7 Passed, 0 Skipped, 0 Expected failures, 0 Unexpected passes, 0 Errors, 0 Reruns 0 Retried, Show all details / Hide all details

Result	Test	Duration	Links
Passed	tests/test_routes.py::test_student_can_submit_hours	113 ms	
Passed	tests/test_routes.py::test_student_can_create_hours_through_json_resource_route	75 ms	
Passed	tests/test_routes.py::test_zero_hours_request_fails_form_validation	61 ms	
Passed	tests/test_routes.py::test_admin_can_review_a_pending_submission	112 ms	
Passed	tests/test_routes.py::test_student_can_redeem_reward	64 ms	
Passed	tests/test_routes.py::test_unauthenticated_admin_request_returns_401	14 ms	
Passed	tests/test_routes.py::test_student_admin_request_returns_403	62 ms	

Postman Collection Run Report




POST Hours resource API - POST /api/hours
http://127.0.0.1:5000/api/hours
201 • 13 ms • 307 B 1
PASS Hours API creates a resource without exposing its proof path
POST Login as admin - POST /login
http://127.0.0.1:5000/login
200 • 8 ms • 6.669 KB 1
PASS Admin login sets the cookie and redirects to the admin page
GET Pending submissions - GET /admin
http://127.0.0.1:5000/admin
200 • 6 ms • 6.669 KB 1
PASS Pending list contains the submitted hours and captures its generated id
GET Review details - GET /admin/hours/:submission_id
http://127.0.0.1:5000/admin/hours/1
200 • 10 ms • 6.926 KB 1
PASS Review detail shows the submitted activity and hours
GET View proof - GET /admin/hours/:submission_id/proof
http://127.0.0.1:5000/admin/hours/1/proof
200 • 49 ms • 469 B 1
PASS Admin proof route returns the uploaded PDF
POST Approve hours - POST /admin/hours/:submission_id/review
http://127.0.0.1:5000/admin/hours/1/review
200 • 6 ms • 6.293 KB 1
PASS Admin approval redirects to the pending list
POST Login as student again - POST /login
http://127.0.0.1:5000/login
200 • 6 ms • 8.198 KB 1
PASS Student login restores the student cookie session



The Newman report shows 19 requests, 19 test scripts, and 19 assertions with zero failures.
The screenshots are arranged in collection order and show requests sent to http://127.0.0.1:5000, including the POST /api/hours resource endpoint.